

Approaches to Supporting Teamwork Solidarity: *Students' Peer Feedback*

- **CONCEPT:** Student teams are not communicating well because they have not properly set expectations on communication, e.g., the modality, frequency (how often), cadence (when), etc. They may also struggle with task coordination (what are their tasks) and uneven workload distribution among the team members.
- **GOALS:** Improve team communication, task coordination, and workload distribution.
- **STUDENT PROMPT OR TASK:**
 - *Communication:*
 - Have students create a deliberate plan for communication.
 - At the end of a sprint or the semester, students could be requested to report what their expectations were for how they were supposed to be communicating, relative to how they actually communicated.
 - *Task Coordination:*
 - Have students create a deliberate plan for task coordination.
 - At the end of a sprint or the semester:
 - Students could be requested to report on the tasks assigned to everyone.
 - Students could be requested to report on the progress of the tasks assigned to everyone: "What is each member of your team assigned to do in this sprint? Additionally, describe each person's progress on their task."
 - *Workload Distribution*
 - Have students create a deliberate plan for workload distribution.
 - At the end of a sprint or the semester:
 - Students could be requested to answer how much each person's work interfaced with their own. Or, if the work was siloed, what did each of their team members work on, and how WILL it interface?
 - This could be reported as a graph of their group work to visualize how items were distributed. This could be color-coded by people, with tasks being nodes. Alternatively, a node could be each member, and a weighted edge would indicate how much each member works together. No edge included would mean that they do not work together.
- **FACULTY ROLE/EXPECTATIONS WITH DESCRIBED TASK:**
 - Have the teams fill out expectations or create a deliberate plan for their group and submit them.

- Have the teams report back or submit a graph of group work (e.g., for workload distribution).
- Potentially use a pop-quiz-style approach to check if the students have a cohesive group and are using the agreed-upon practices.
- **IMPLEMENTATION WINDOW** (*When in the semester is this used?*):
 - Teams could submit their expectations for communication, task coordination, and/or workload distribution at the beginning of the semester or of a sprint.
 - Teams will report how it went at the end of a sprint or semester.
 - During the semester, the pop quiz-style check-ins could be conducted in class, without a lot of preparation and without access to the team's previous submission of expectations. By asking each group member to answer each question without giving them time to consult with each other, it can be easier to see if they have clarity in their plans, if the solidarity is really there, and to determine whether they are working as a cohesive group.
- **FACULTY EVALUATION CRITERIA SUGGESTED:**
 - Faculty can compare group members' submitted responses.
 - For the pop-quiz-style check-ins, each group member could be asked to answer each question without giving the students time to consult with each other.
 - Assessment: Responses could be evaluated by looking at how much each team member knows about the coordination but also how similar the different answers are (which can measure coordination) using the following grading criteria:
 - 0 - No answer or no response given
 - 1 - Got a lot of things wrong
 - 2 - Mostly correct, but may have gotten one or two things wrong
 - 3 - Accurately reported the progress of each team member
 - In the case of assessing a graph submitted to see the *workload distribution*: It could be considered a bad sign if the groups build "trivial graphs" where everyone is connected and has the same weight. Or, if any group has a member that is truly disconnected from multiple group members' drawings.
- **POTENTIAL CONCERNS AND APPROACHES:**
 - **PROBLEM:** Challenges for faculty to compare students' responses.
 - **SOLUTIONS:**
 - Have students submit something in a Google Form (or Microsoft or another tool) where it is easy for faculty to write a script to analyze the group output/consistency.

- Create some system that allows the group to see the responses from each team member with the identity removed. This allows the team to also see how cohesive their responses are, which can help them reflect on their team communication strengths and pitfalls.